

SATVIK AITHARAJU

AI Engineer / Data Engineer

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Professional Summary

AI Engineer with 1 year of experience designing and shipping production LLM applications, RAG pipelines, and multi-agent systems. Proficient in Python, FastAPI, LangChain, and LangGraph, with hands-on experience in Agentic AI and the Model Context Protocol (MCP). Skilled in cloud-native AI development on AWS (S3, Lambda, Bedrock), prompt engineering, and building scalable data pipelines that turn LLMs into reliable, business-ready solutions.

Education

University of North Texas

Master of Science in Information Systems and Technology

Aug 2024 – May 2026

GPA: 3.7 / 4.0

Professional Experience

AI Data Engineer

Dell Technologies

May 2023 – Apr 2024

- Designed and deployed scalable ETL pipelines using Python, PySpark, Apache Spark, Databricks, AWS S3, and Lambda, reducing pipeline execution time by 35% and improving enterprise AI data processing.
- Built an AI-powered Document Intelligence System using semantic search and RAG, reducing information retrieval time by 40% and improving response accuracy for engineering teams.
- Fine-tuned GPT, BERT, and T5 models using Hugging Face and TensorFlow, integrating RAG and prompt engineering to improve AI response accuracy by 30%.
- Collaborated with cross-functional teams to gather requirements, translate business needs into technical AI solutions, and deliver scalable backend services for enterprise AI and data applications.
- Integrated data pipelines with MLOps workflows using MLflow, Docker, Kubernetes, GitHub Actions, and CI/CD pipelines, enabling automated model training, deployment, monitoring, and operational reliability.

Projects

Agentic AI Health Assistant using MCP and LLMs

FastAPI, LangChain, LangGraph, OpenAI, MCP

- Developed an AI-powered healthcare assistant using OpenAI, FastAPI, LangChain, LangGraph, and MCP, implementing AI agents, prompt engineering, and dynamic tool calling to deliver context-aware health guidance and symptom analysis.
- Designed Retrieval-Augmented Generation (RAG) workflows with multi-agent orchestration, MCP servers, and external APIs, incorporating unit testing and validation checks to improve response accuracy, reliability, and production readiness.

AI-Powered Document Intelligence System

Python, FastAPI, LangChain, OpenAI, FAISS, Azure, Databricks

- Developed an intelligent document processing and Retrieval-Augmented Generation (RAG) platform that ingested technical manuals, engineering documents, and knowledge base articles, enabling semantic search and context-aware answering.
- Built scalable data ingestion, embedding, & vector indexing pipelines using PySpark, Databricks, FAISS, and Azure services, integrating LLMs to provide low-latency insights and improve information retrieval for engineering & support teams.

Technical Skills

Programming Languages: Python, SQL, HTML, R, C++, MySQL

Machine Learning: Supervised & Unsupervised Learning, Deep Learning, NLP, Computer Vision, Time Series Analysis, Model Evaluation

Generative AI & LLMs: Prompt Engineering, Prompt Templates, Retrieval-Augmented Generation (RAG), LLM Fine-tuning, AI Agents, Agentic AI, Multi-Agent Systems, Agent Orchestration, Function Calling, Tool Calling, MCP

Frameworks & Libraries: FastAPI, LangChain, LangGraph, LlamaIndex, PyTorch, TensorFlow, Keras, Hugging Face

Vector Databases: FAISS, ChromaDB, Pinecone, Weaviate

Cloud & Data Engineering: AWS, AWS S3, AWS Lambda, AWS Bedrock, Azure, Apache Spark, PySpark, Databricks, Kafka, Hadoop, Airflow, ETL Pipelines, Snowflake

MLOps & DevOps: Docker, Kubernetes, MLflow, Git, GitHub Actions, CI/CD, REST APIs, Unit Testing, Backend Services

Collaboration: Requirements Gathering, Cross-functional Collaboration, Technical Documentation, Agile Development